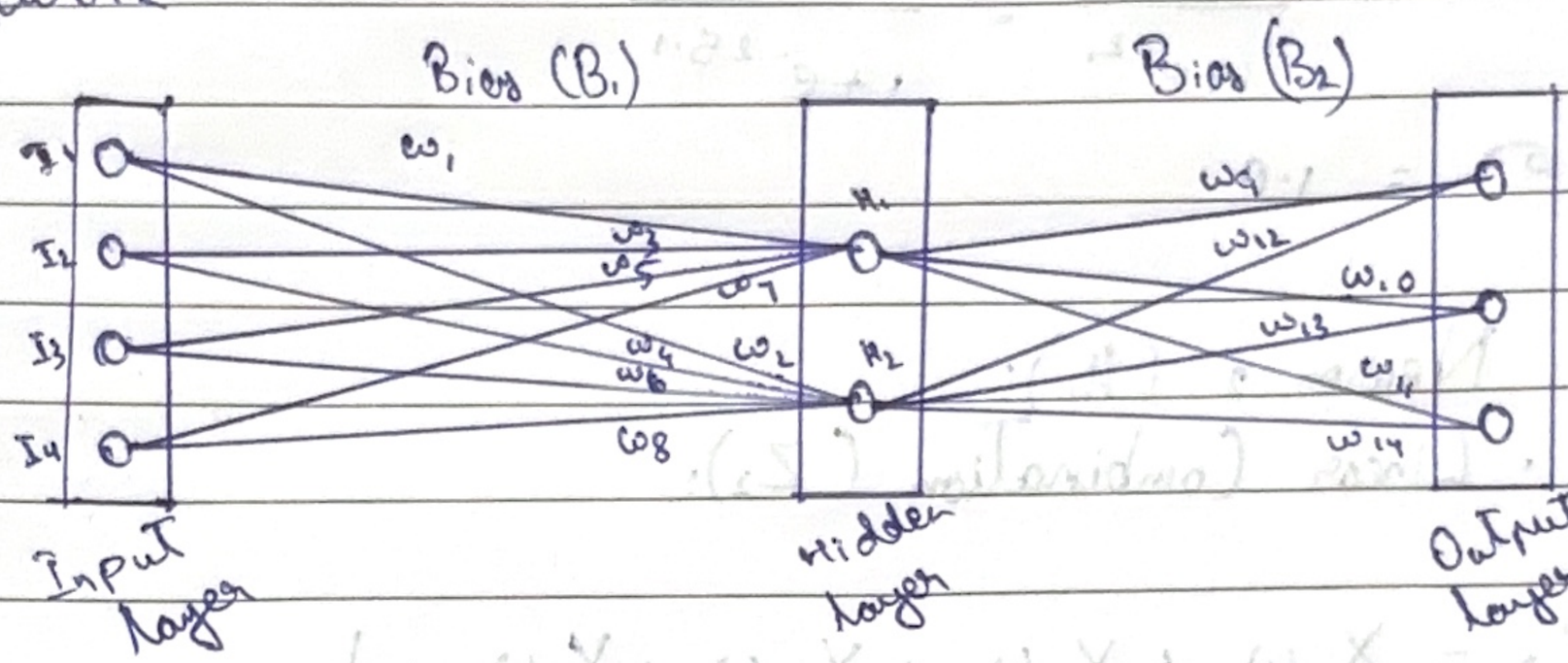
Mathematical Calculation Single - Hidden - Layer Neural Network



1 Given Parameters

Input Vectors : $X = [5.1, 3.2, 1.4, 0.2]$

- Hidden weights (H_1) : $\omega_1 = 1$ $\omega_3 = 3$ $\omega_5 = 5$ $\omega_7 = 7$
- ~~$\omega_5 = 5$~~ Hidden weights (H_2) : $\omega_2 = 2$, $\omega_4 = 4$, $\omega_6 = 6$, $\omega_8 = 8$
- ~~Hidden weights (H_2)~~ Output weights (from H_2) : $\omega_9 = 0.5$ 1.3 $\omega_{10} = 0.5$ $\omega_{11} = 1.2$
- Output weights (from H_2) : $\omega_{12} = 0.5$ $\omega_{13} = 0.2$ $\omega_{14} = 0.4$
- Biases : $b_1 = 2$ $b_2 = 0.1$

2 Hidden Layer Calculation

A) Neuron 1 (H_1) :

- Linear Combination (Z_1) :

$$Z_1 = X_1\omega_1 + X_2\omega_3 + X_3\omega_3 + X_4\omega_7 + b_1$$

$$Z_1 = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2$$

$$Z_1 = 5.1 + 9.6 + 7 + 1.4 \times 2$$

$$Z_1 = 25.1$$

. Activation (σ_1):

$$\sigma_1 = \frac{1}{1+e^{-Z_1}} = \frac{1}{1+e^{-25.1}}$$

$$\sigma_1 = 1.0$$

B) Neuron 2 (H_2):

. Linear Combination (Z_2):

$$Z_2 = X_1 w_2 + X_2 w_4 + X_3 w_6 + X_4 w_8 + b_1$$
$$Z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) + 2$$
$$Z_2 = \underline{35.0}$$

. Activation (σ_2):

$$\sigma_2 = \frac{1}{1+e^{-Z_2}} = \frac{1}{1+e^{-35.0}}$$

$$\sigma = 1.0$$

3 Output Layer Calculation (combining both neurons)

A) Setosa (Z setosa):

$$Z_{setosa} = (\sigma_1 \times w_9) + (\sigma_2 \times w_{12}) + b_2$$
$$= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1$$
$$= 1.3 + 0.5 + 0.1$$
$$Z_{sesota} = 1.9$$

B) Versicolor (Z versicolor):

$$Z_{versicolor} = (\sigma_1 \times w_{10}) + (\sigma_2 \times w_{13}) + b_2$$
$$= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1$$
$$= 0.5 + 0.2 + 0.1$$
$$Z_{versicolor} = 0.8$$

c) ~~Verginicia~~ = Verginicia (Z verginicia) :

$$Z_{verginicia} = (\sigma_1 \times w_1) + (\sigma_2 \times w_{12}) + b_2$$
$$= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1$$
$$= 1.2 + 0.4 + 0.1$$
$$Z_{verginicia} = 1.7$$

a) Final ~~Output~~ Prediction

- Output vector : [1.9, 0.8, 1.7]
- Prediction : setosa [Highest sure 1.9]